

LECTURE-7

plasma membrane



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

7

Membrane Structure and Function

Key concepts: Control of molecular traffic by plasma membrane



- ❖ The plasma membrane separates the living cell from its surroundings.
- ❖ This thin barrier, 8 nm thick, controls traffic into and out of the cell.
- ❖ Like all biological membranes, the plasma membrane is **selectively permeable**, allowing some substances to cross more easily than others.
- ❖ The formation of a membrane that encloses a solution different from the surrounding solution while still permitting *the uptake of nutrients and the elimination of waste products* was a **key event in the evolution of life**.
- ❖ The **ability of the cell to discriminate in its chemical exchanges with its environment** is fundamental to life.
- ❖ It is the **plasma membrane and its component molecules** that **make this selectivity possible**.

KEY CONCEPTS

- 7.1 Cellular membranes are fluid mosaics of lipids and proteins
- 7.2 Membrane structure results in selective permeability
- 7.3 Passive transport is diffusion of a substance across a membrane with no energy investment
- 7.4 Active transport uses energy to move solutes against their gradients
- 7.5 Bulk transport across the plasma membrane occurs by exocytosis and endocytosis

Membranes are fluid: Lipid flip-flops



- ❖ Membrane molecules are **held in place by relatively weak hydrophobic interactions**.
- ❖ Most of the **lipids and some proteins drift laterally** in the plane of the membrane but **rarely flip-flop** from one phospholipid layer to the other.
 - ✓ Adjacent phospholipids switch positions about 10^7 times per second.
 - ✓ The lateral movements of phospholipids are rapid, about 2 μm per second.
 - ✓ A phospholipid can travel the length of a typical bacterial cell in 1 sec.

Membranes are fluid: Lipid flip-flops



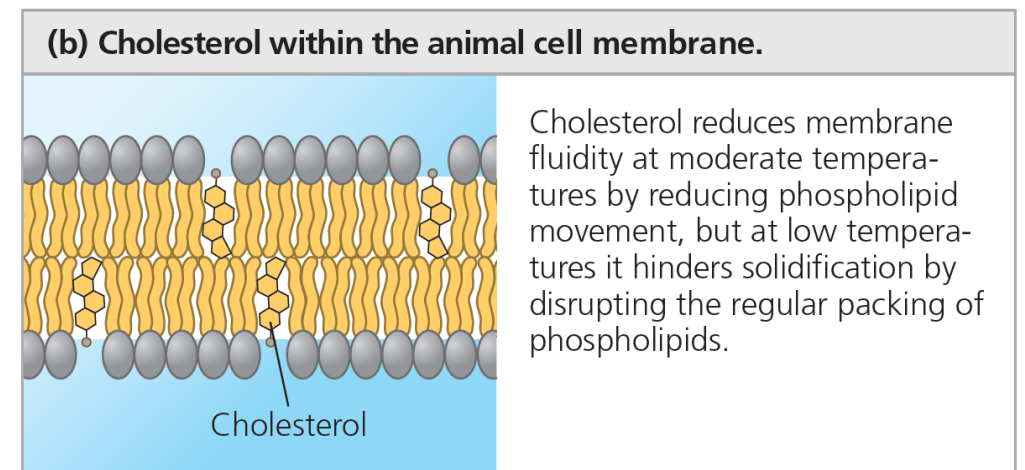
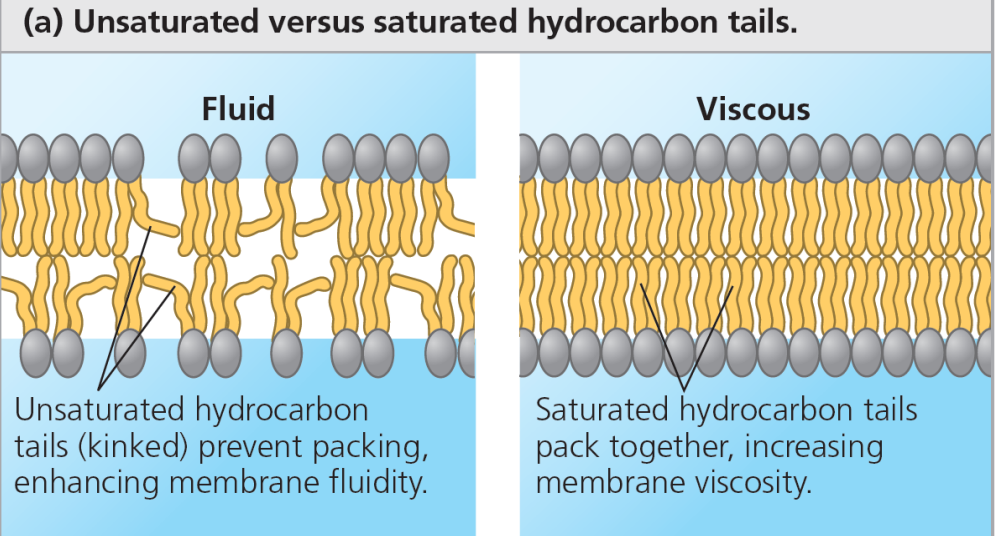
- ❖ Some large membrane **proteins drift** within the phospholipid bilayer, although they move more slowly than the phospholipids.
- ❖ **Some proteins move in a very directed manner**, perhaps guided or driven by motor proteins attached to the cytoskeleton.
- ❖ **Other proteins never move** and are anchored to the cytoskeleton or to the extracellular matrix.
- ❖ Membrane **fluidity is influenced by temperature**.
 - ❖ As temperatures cool, membranes switch from a fluid state to a solid state as the phospholipids pack more closely.

Fluidity in membranes



- ❖ Membrane fluidity is **also influenced by the components of the membrane**.
 - ✓ *Membranes rich in unsaturated fatty acids are more fluid than those dominated by saturated fatty acids because kinks in the unsaturated fatty acid tails at the locations of the double bonds prevent tight packing.*
- ❖ The **steroid cholesterol is wedged between phospholipid molecules** in the plasma membrane of **animal cells**.
 - ✓ *At warm temperatures (such as 37°C), cholesterol restrains the movement of phospholipids and reduces fluidity.*
 - ✓ *At cool temperatures, cholesterol maintains fluidity by preventing tight packing.*
 - ✓ *Thus, **cholesterol acts as a “fluidity buffer”** for the membrane, resisting changes in membrane fluidity as temperature changes.*

▼ **Figure 7.5** Factors that affect membrane fluidity.

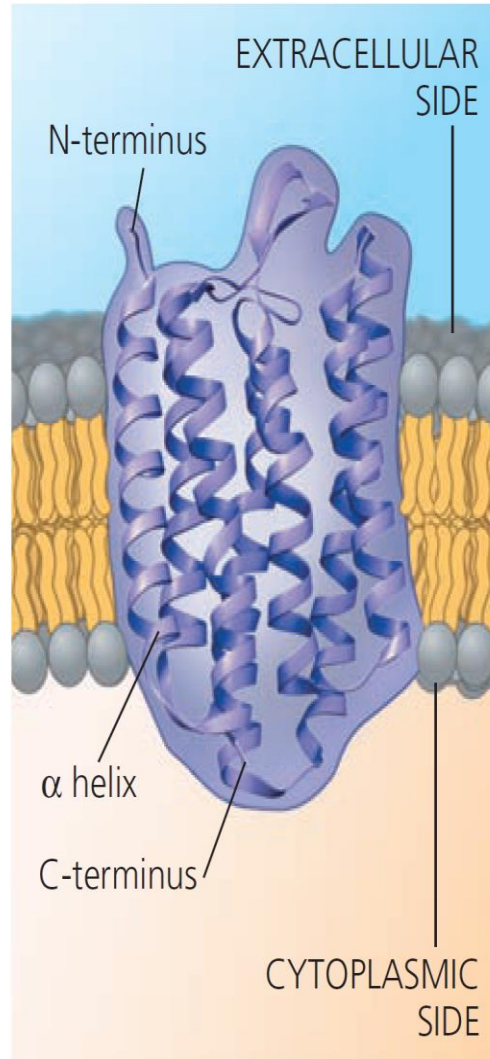


Factors affect fluidity and protein composition in lipids



► **Figure 7.6** The structure of a transmembrane protein.

Bacteriorhodopsin (a bacterial transport protein) has a distinct orientation in the membrane, with its N-terminus outside the cell and its C-terminus inside. This ribbon model highlights the α -helical secondary structure of the hydrophobic parts, which lie mostly within the hydrophobic interior of the membrane. The protein includes seven transmembrane helices. The nonhelical hydrophilic segments are in contact with the aqueous solutions on the extracellular and cytoplasmic sides of the membrane. Although shown as simple purple shapes in many figures in this book, each protein has its own unique structure.



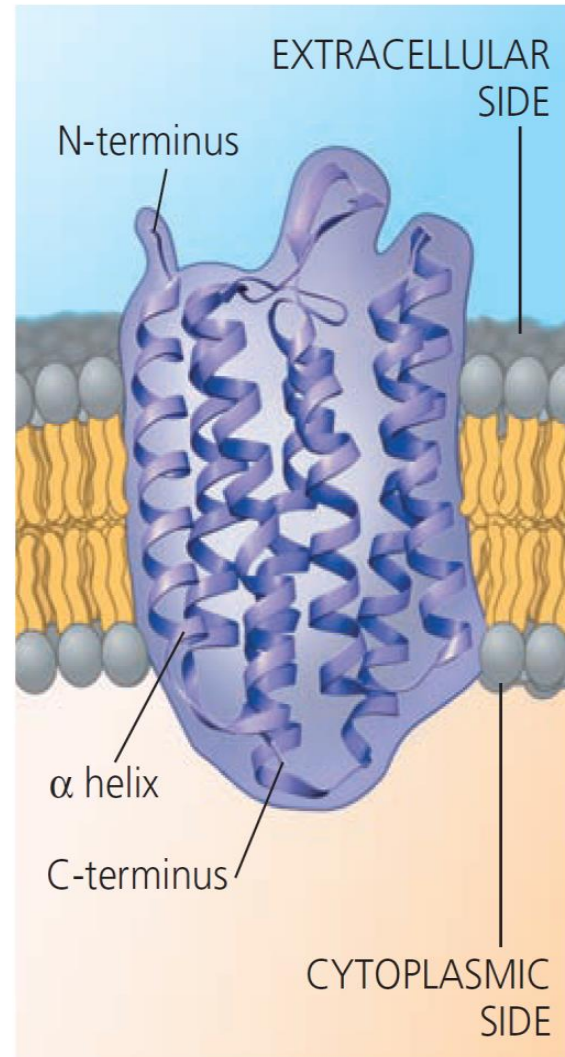
- ❖ There are two major populations of **membrane proteins**: **integral and peripheral**.
- ❖ **Integral proteins** penetrate the hydrophobic interior of the lipid bilayer, usually completely spanning the membrane as *transmembrane proteins*.
 - ✓ Other integral proteins extend partway into the hydrophobic interior.
 - ✓ The **hydrophobic regions** embedded in the membrane's interior **consist of stretches of nonpolar amino acids, usually coiled into alpha helices**.
 - ✓ The hydrophilic regions of integral proteins are in contact with the aqueous environment.
 - ✓ Some integral proteins have a **hydrophilic channel** through their center that **allows passage of hydrophilic substances**.

Transmembrane and Peripheral proteins in membrane



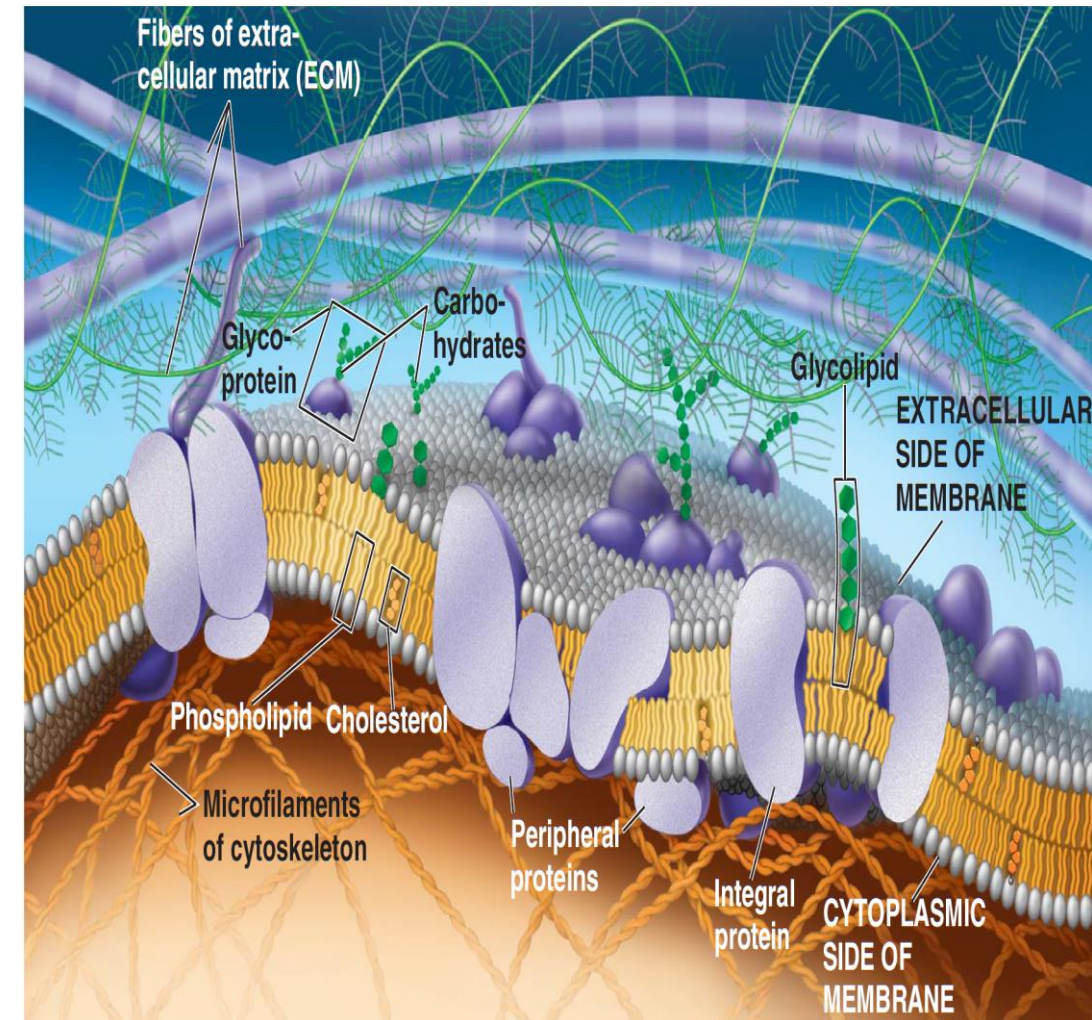
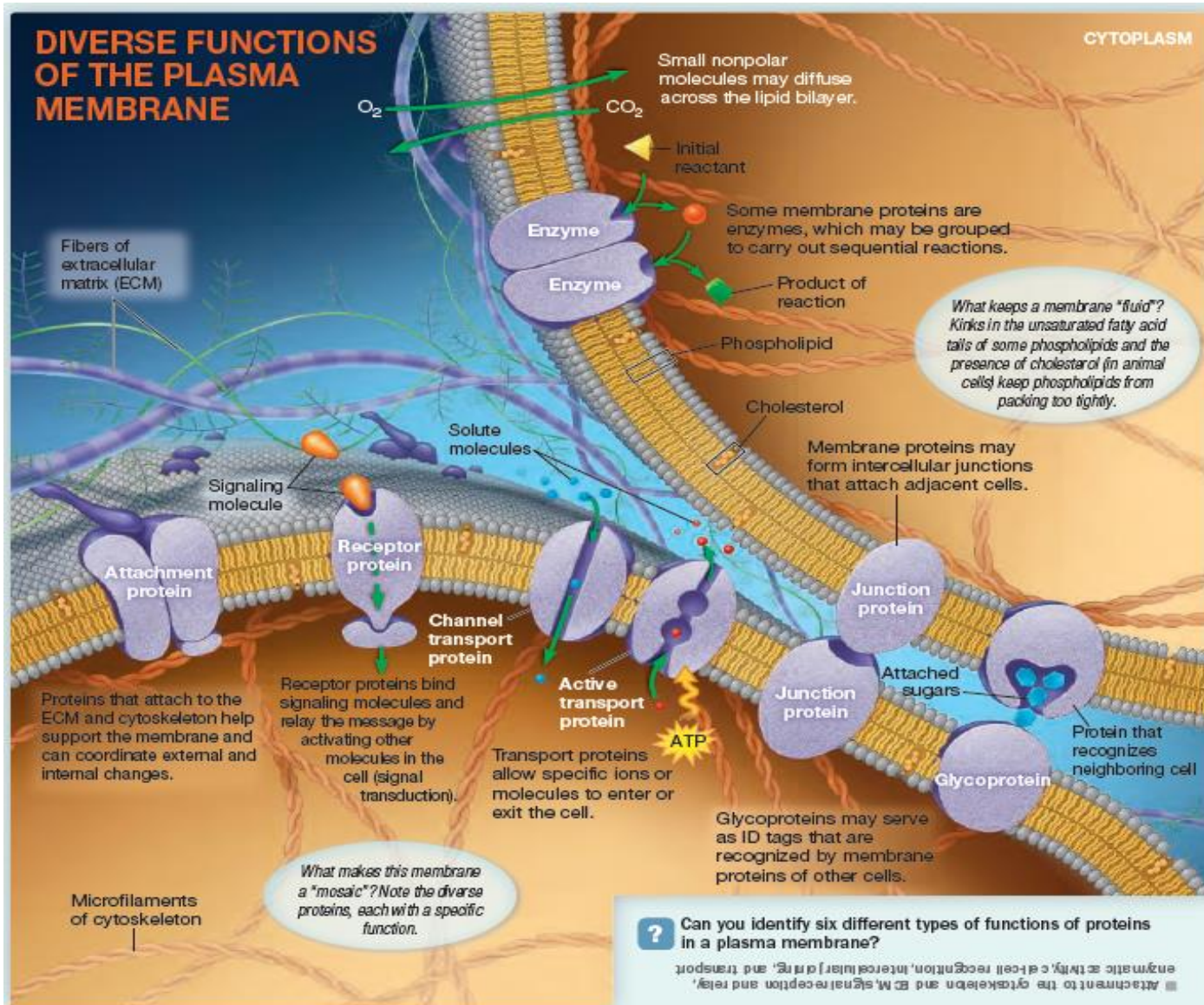
► **Figure 7.6** The structure of a transmembrane protein.

Bacteriorhodopsin (a bacterial transport protein) has a distinct orientation in the membrane, with its N-terminus outside the cell and its C-terminus inside. This ribbon model highlights the α -helical secondary structure of the hydrophobic parts, which lie mostly within the hydrophobic interior of the membrane. The protein includes seven transmembrane helices. The nonhelical hydrophilic segments are in contact with the aqueous solutions on the extracellular and cytoplasmic sides of the membrane. Although shown as simple purple shapes in many figures in this book, each protein has its own unique structure.



- ❖ **Peripheral proteins** are not embedded in the lipid bilayer at all.
 - ✓ Instead, peripheral proteins are *loosely bound to the surface of the membrane*, often to integral proteins.
- ❖ *On the cytoplasmic side* of the membrane, some membrane proteins are *attached to the cytoskeleton*.
- ❖ *On the extracellular side* of the membrane, some membrane proteins *attach to the fibers of the extracellular matrix*.
- ❖ **These attachments combine to give animal cells a stronger framework than the plasma membrane itself could provide.**

Diverse function of plasma membrane



Structure and function of membrane mosaics



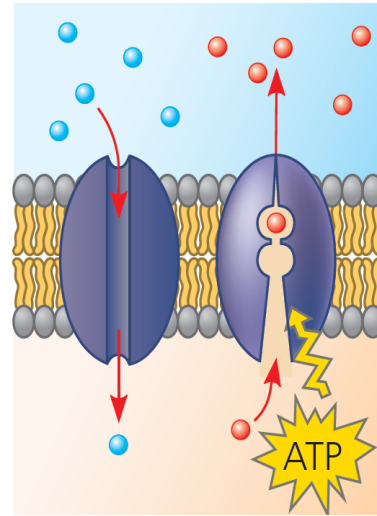
❖ The **proteins of the plasma membrane** have **six major functions**:

1. **Transport** of specific solutes into or out of cells
2. **Enzymatic activity**, sometimes catalyzing one of a number of steps of a metabolic pathway
3. **Signal transduction**, relaying hormonal messages to the cell
4. **Cell-cell recognition**, allowing other proteins to attach two adjacent cells together
5. **Intercellular joining** of adjacent cells with gap or tight junctions
6. Attachment to the cytoskeleton and extracellular matrix, **maintaining cell shape and stabilizing the location** of certain membrane proteins

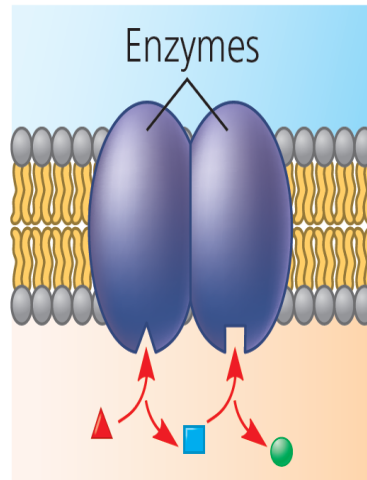
❖ Proteins on the surface of a cell may help outside agents invade the cell.

- ✓ *The human immunodeficiency virus (HIV) infects immune system cells by binding to cell surface proteins.*

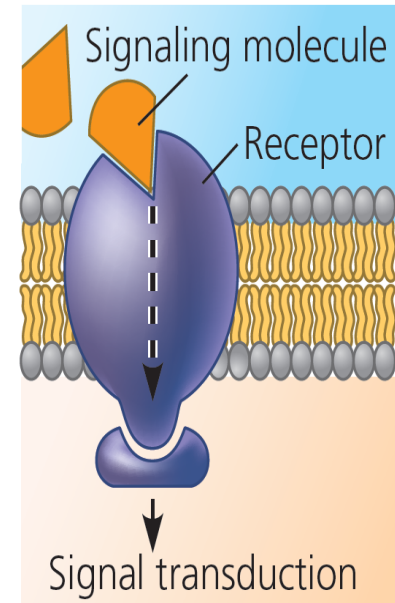
(a) Transport. *Left:* A protein that spans the membrane may provide a hydrophilic channel across the membrane that is selective for a particular solute. *Right:* Other transport proteins shuttle a substance from one side to the other by changing shape (see Figure 7.14b). Some of these proteins hydrolyze ATP as an energy source to actively pump substances across the membrane.



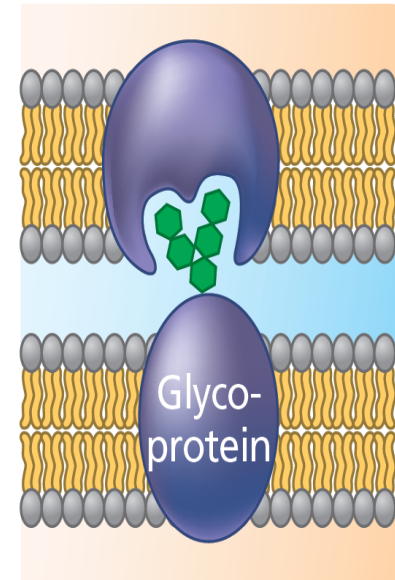
(b) Enzymatic activity. A protein built into the membrane may be an enzyme with its active site exposed to substances in the adjacent solution. In some cases, several enzymes in a membrane are organized as a team that carries out sequential steps of a metabolic pathway.



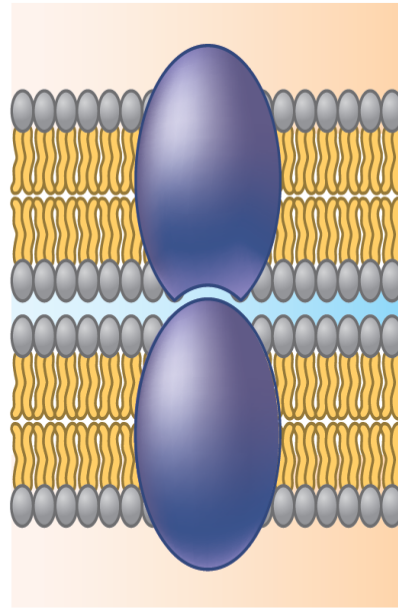
(c) Signal transduction. A membrane protein (receptor) may have a binding site with a specific shape that fits the shape of a chemical messenger, such as a hormone. The external messenger (signaling molecule) may cause the protein to change shape, allowing it to relay the message to the inside of the cell, usually by binding to a cytoplasmic protein (see Figure 11.6).



(d) Cell-cell recognition. Some glycoproteins serve as identification tags that are specifically recognized by membrane proteins of other cells. This type of cell-cell binding is usually short-lived compared to that shown in (e).

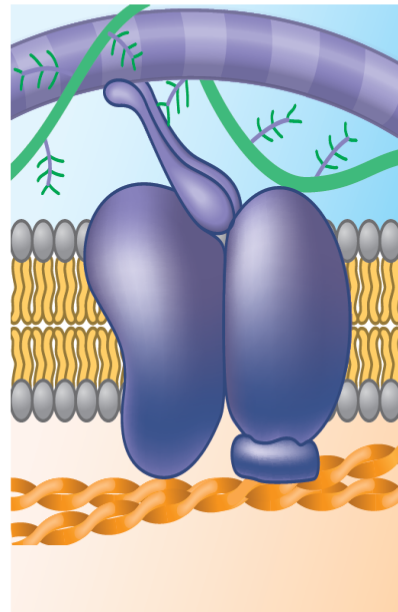


(e) Intercellular joining. Membrane proteins of adjacent cells may hook together in various kinds of junctions, such as gap junctions or tight junctions (see Figure 6.30). This type of binding is more long-lasting than that shown in (d).



(f) Attachment to the cytoskeleton and extracellular matrix (ECM).

Microfilaments or other elements of the cytoskeleton may be noncovalently bound to membrane proteins, a function that helps maintain cell shape and stabilizes the location of certain membrane proteins. Proteins that can bind to ECM molecules can coordinate extracellular and intracellular changes (see Figure 6.28).



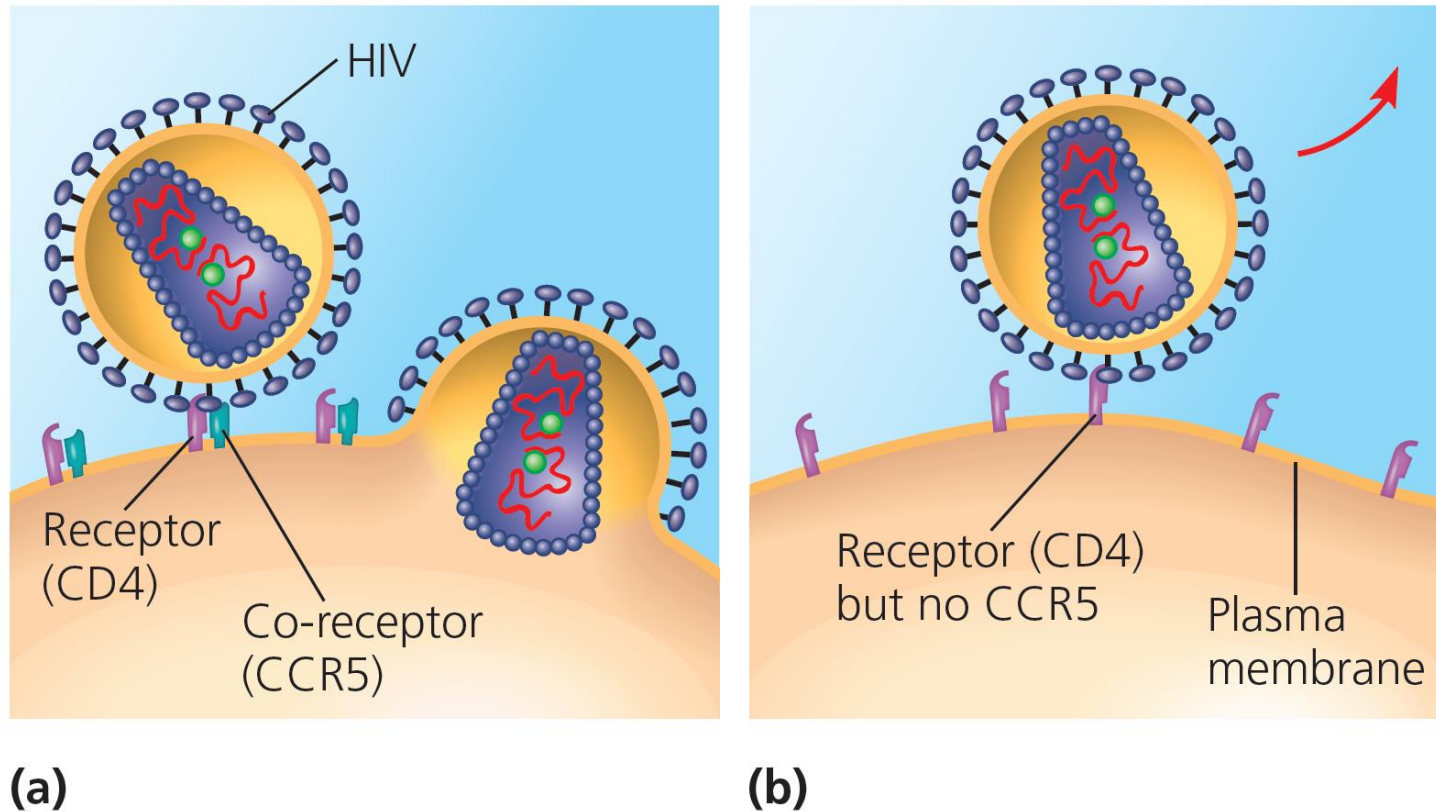
❖ **Cell-cell recognition**, the ability of a cell to distinguish one type of neighboring cell from another, is crucial to the functioning of an organism.

- ✓ *Cell-cell recognition is important in the sorting and organizing of cells into tissues and organs during development.*
- ✓ *Recognition is also the basis for the rejection of foreign cells by the immune system.*
- ✓ *Cells recognize other cells by binding to surface molecules, often containing carbohydrates, on the extracellular surface of the plasma membrane.*

Carbohydrate is important for cell-cell recognition

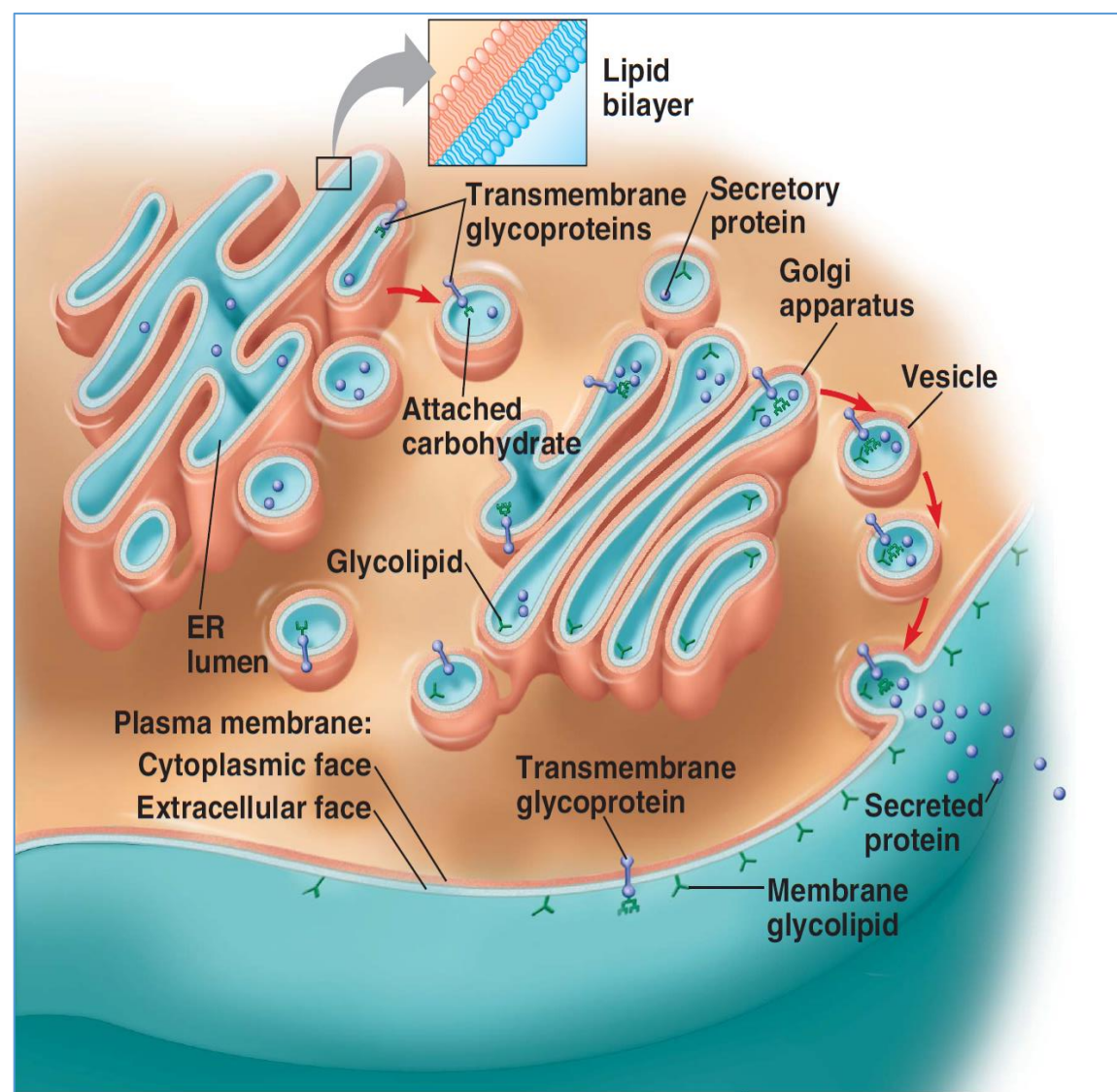
- ❖ Membrane carbohydrates are usually branched chains with fewer than 15 sugar units.
- ❖ Membrane carbohydrates may be covalently bonded to lipids, forming **glycolipids**, or more commonly to proteins, forming **glycoproteins**.
- ❖ The carbohydrates on the extracellular side of the plasma membrane *vary from species to species, from individual to individual, and even from cell type to cell type within an individual*.
 - ✓ **The four human blood groups (A, B, AB, and O) differ in the carbohydrate part of glycoproteins on the surface of red blood cells.**

Genetic basis for HIV resistance



▲ **Figure 7.8 The genetic basis for HIV resistance.** (a) HIV can infect a cell with CCR5 on its surface, as in most people. (b) HIV cannot infect a cell lacking CCR5 on its surface, as in resistant individuals.

Membranes have distinct inside and outside faces: Role of ER and GA



- ❖ The inside and outside faces of membranes *may differ in lipid composition*.
 - ❖ Each protein in the membrane has a *directional orientation in the membrane*.
 - ❖ The asymmetrical arrangement of proteins, lipids, and their associated carbohydrates in the plasma membrane is determined as the membrane is built by the endoplasmic reticulum (ER) and Golgi apparatus.
-
- ✓ *Membrane lipids and proteins are synthesized in the ER.*
 - ✓ *Carbohydrates are added to proteins in the ER, and the resulting glycoproteins are further modified in the Golgi apparatus.*
 - ✓ *Glycolipids are also produced in the Golgi apparatus*

Membrane structure result in selective permeability

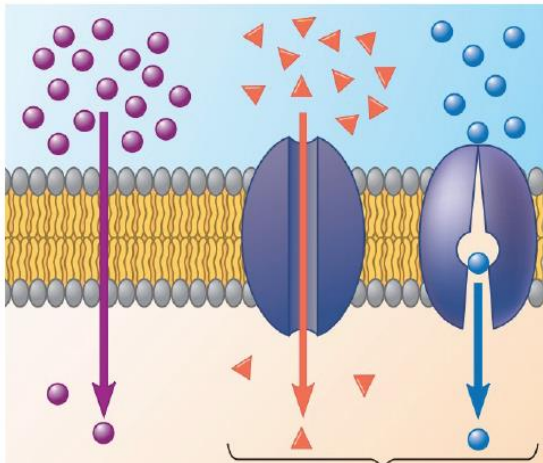
- ❖ ***Movement of a molecule through a membrane*** depends on the interaction of the molecule with the hydrophobic interior of the membrane.
 - ✓ ***Nonpolar molecules***, such as hydrocarbons, CO_2 , and O_2 , are hydrophobic and can dissolve in the lipid bilayer and cross easily, without the assistance of membrane proteins.
 - ✓ The hydrophobic interior of the membrane impedes the direct passage of ions and polar molecules, which are hydrophilic.
 - ✓ ***Polar molecules***, such as glucose and other sugars, and even water, an extremely small polar molecule, cross the lipid bilayer slowly.
 - ✓ An ion, whether a charged atom or a molecule, and its surrounding shell of water also have difficulty penetrating the hydrophobic interior of the membrane.
- ❖ **Proteins assist and regulate the transport of ions and polar molecules.**
- ❖ Cell membranes are permeable to specific ions and a variety of polar molecules, which can avoid contact with the lipid bilayer by passing through **transport proteins** that span the membrane.
- ❖ Some transport proteins called **channel proteins** have a hydrophilic channel that certain molecules or ions can use as a tunnel through the membrane.

Passive and active transport



▼ Figure 7.16 Review: passive and active transport.

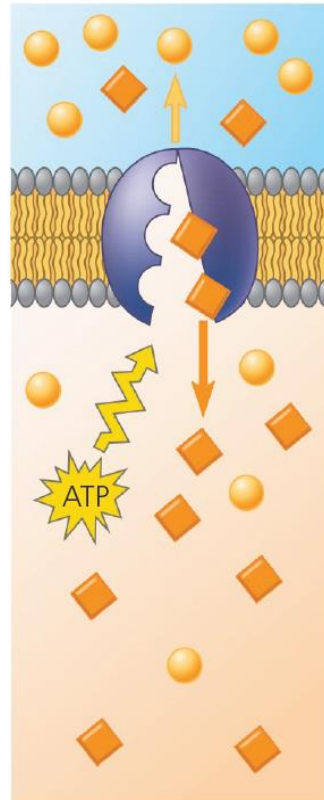
Passive transport. Substances diffuse spontaneously down their concentration gradients, crossing a membrane with no expenditure of energy by the cell. The rate of diffusion can be greatly increased by transport proteins in the membrane.



Diffusion. Hydrophobic molecules and (at a slow rate) very small uncharged polar molecules can diffuse through the lipid bilayer.

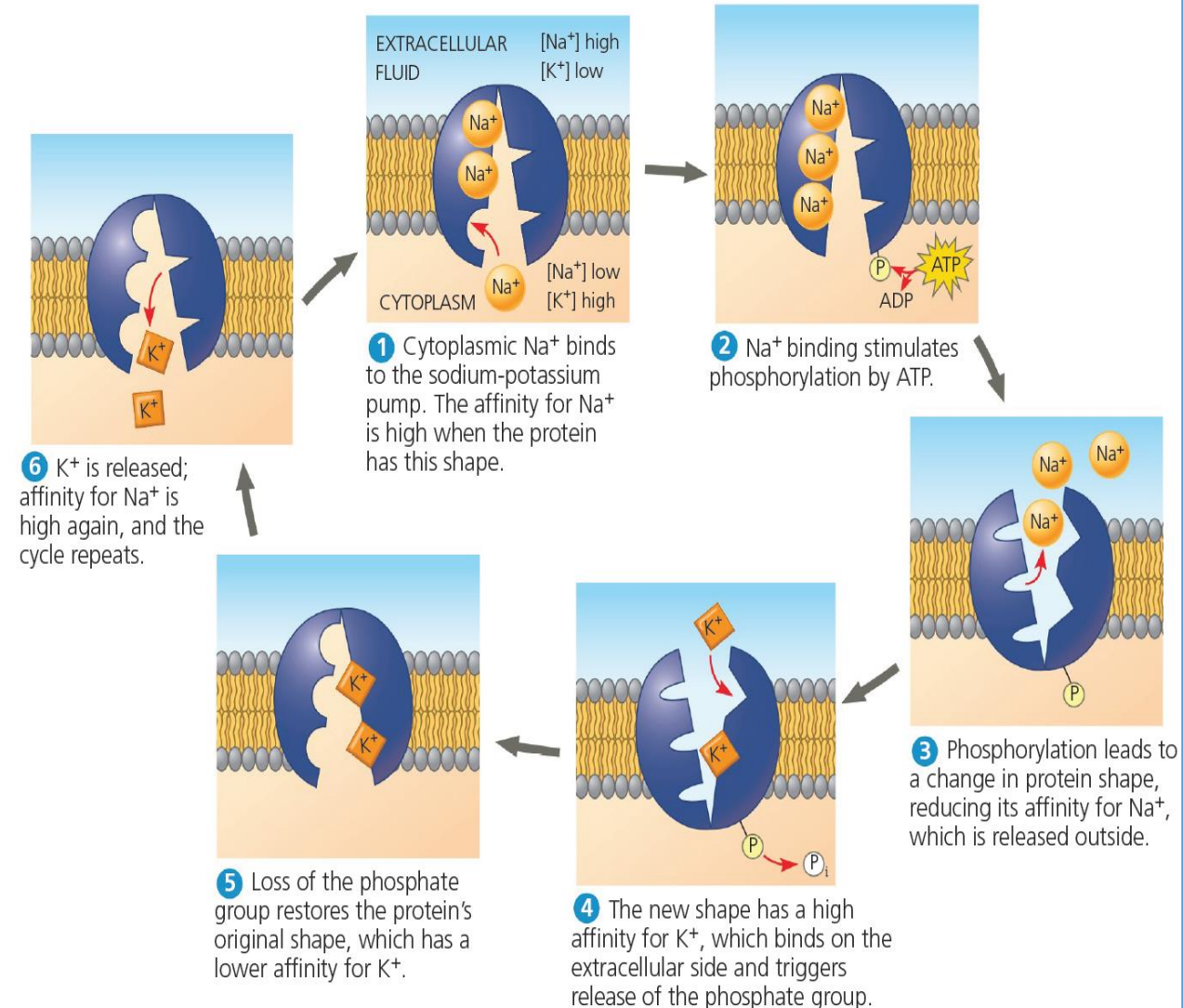
Facilitated diffusion. Many hydrophilic substances diffuse through membranes with the assistance of transport proteins, either channel proteins (left) or carrier proteins (right).

Active transport. Some transport proteins act as pumps, moving substances across a membrane against their concentration (or electrochemical) gradients. Energy for this work is usually supplied by ATP.



► Figure 7.15 The sodium-potassium pump: a specific case of active transport.

This transport system pumps ions against steep concentration gradients: Sodium ion concentration ($[Na^+]$) is high outside the cell and low inside, while potassium ion concentration ($[K^+]$) is low outside the cell and high inside. The pump oscillates between two shapes in a cycle that moves 3 Na^+ out of the cell for every 2 K^+ pumped into the cell. The two shapes have different affinities for Na^+ and K^+ . ATP powers the shape change by transferring a phosphate group to the transport protein (phosphorylating the protein).

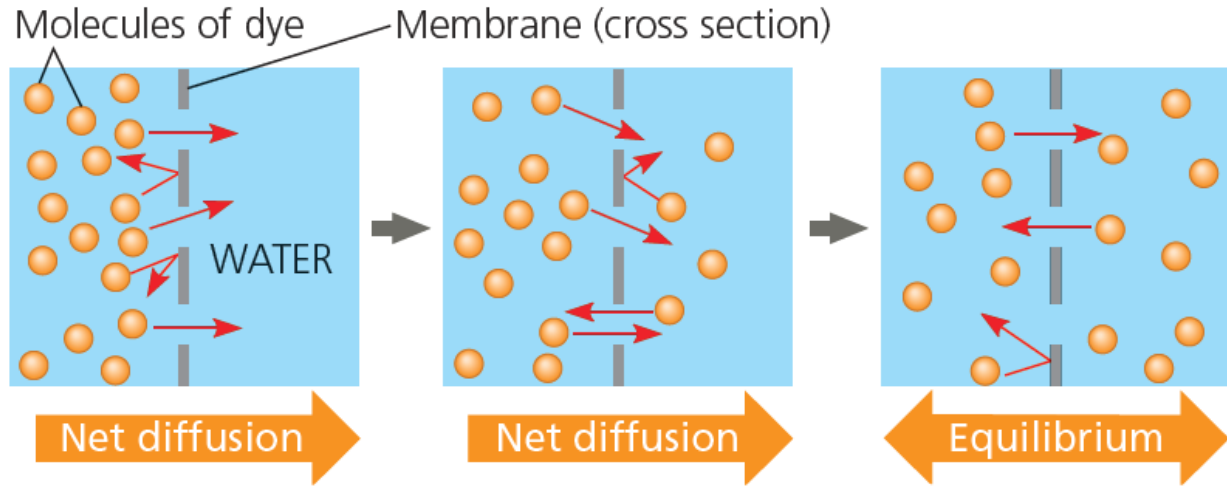


Passive transport due to thermal energy

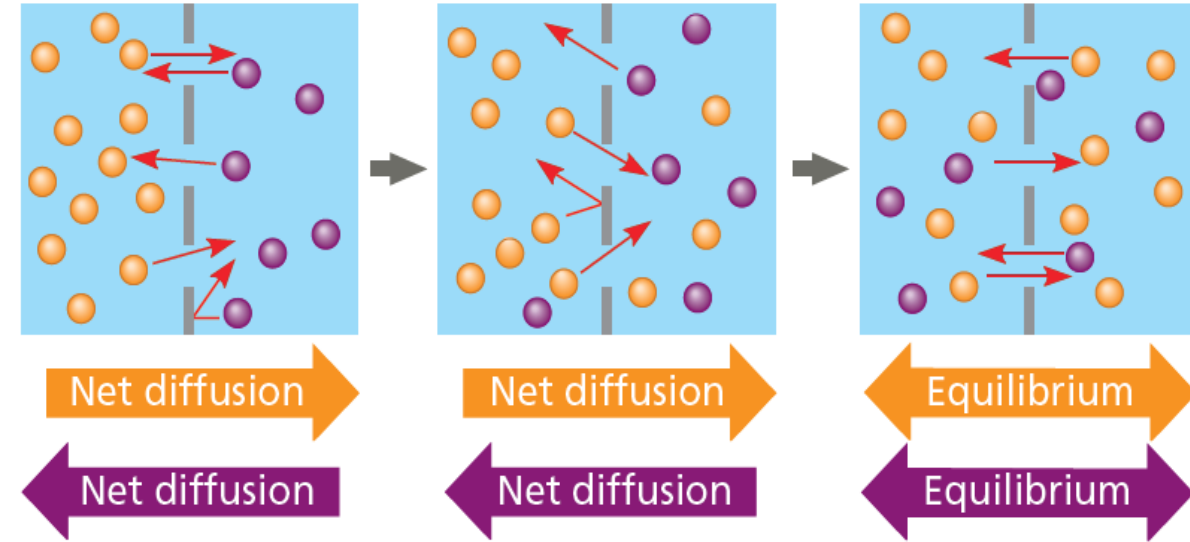


- ❖ Molecules have thermal energy or heat, due to their constant motion.
 - ✓ One result of thermal motion is **diffusion**, the movement of molecules of any substance to spread out in the available space.
- ❖ No work must be done to move substances down the concentration gradient; diffusion is a spontaneous process, needing no input of energy.
- ❖ Each substance diffuses down its *own* concentration gradient, independent of the concentration gradients of other substances.
- ❖ *The diffusion of a substance across a biological membrane is **passive transport** because it requires no energy from the cell to make it happen.*
 - ✓ The concentration gradient itself represents potential energy and drives diffusion. ex., aquaporins.
- ❖ Because membranes are selectively permeable, the interactions of the molecules with the membrane play a role in the diffusion rate.

Diffusion of solutes

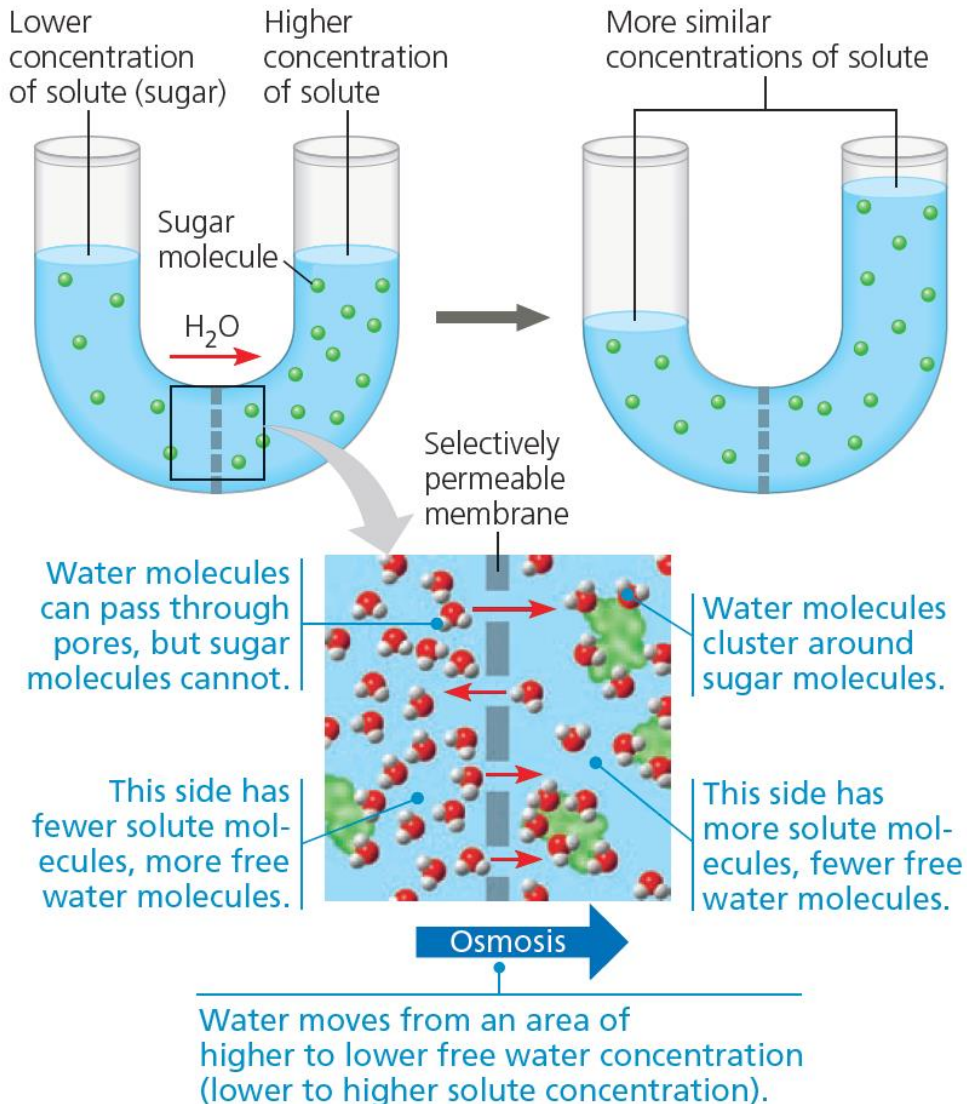


(a) Diffusion of one solute. The membrane has pores large enough for molecules of dye to pass through. Random movement of dye molecules will cause some to pass through the pores; this will happen more often on the side with more dye molecules. The dye diffuses from where it is more concentrated to where it is less concentrated (called diffusing down a concentration gradient). This leads to a dynamic equilibrium: The solute molecules continue to cross the membrane, but at roughly equal rates in both directions.



(b) Diffusion of two solutes. Solutions of two different dyes are separated by a membrane that is permeable to both. Each dye diffuses down its own concentration gradient. There will be a net diffusion of the purple dye toward the left, even though the *total* solute concentration was initially greater on the left side.

Diffusion of solvent



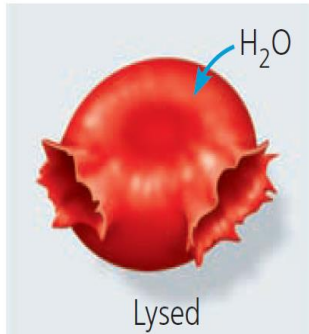
- ❖ **Osmosis.** Two sugar solutions of different concentrations are separated by a membrane that the solvent (water) can pass through but the solute (sugar) cannot.
- ❖ Water molecules move randomly and may cross in either direction, but overall, water diffuses from the solution with less concentrated solute to that with more concentrated solute.
- ❖ This passive transport of water, or osmosis, makes the sugar concentrations on both sides more nearly equal.

Osmosis-example



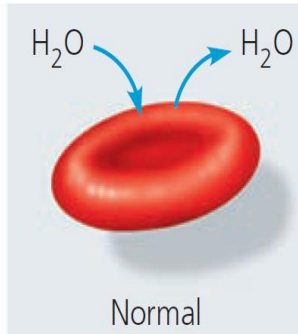
(a) Animal cell. An animal cell fares best in an isotonic environment unless it has special adaptations that offset the osmotic uptake or loss of water.

Hypotonic solution



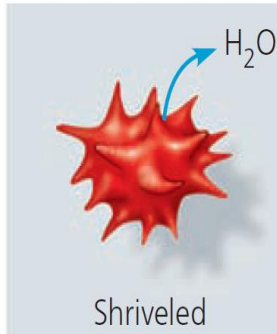
Lysed

Isotonic solution



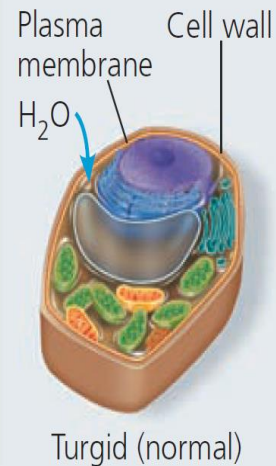
Normal

Hypertonic solution

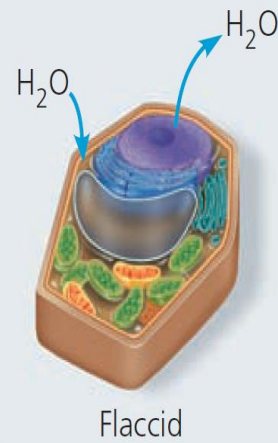


Shriveled

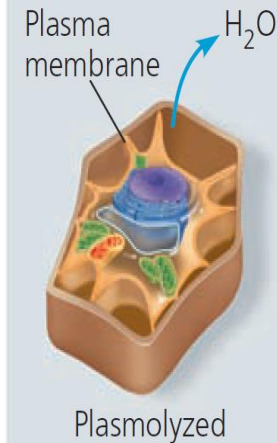
(b) Plant cell. Plant cells are turgid (firm) and generally healthiest in a hypotonic environment, where the uptake of water is eventually balanced by the wall pushing back on the cell.



Turgid (normal)



Flaccid



Plasmolyzed

- ❖ If the cell is immersed in a solution that is **hypertonic** to the cell (containing nonpenetrating solutes), the cell loses water to its environment, shrivels, and probably dies.
 - ✓ *For example, an increase in the salinity (saltiness) of a lake can kill aquatic animals.*
 - ✓ *If the lake water becomes hypertonic to the animals' cells, the cells may shrivel and die.*
- ❖ Taking up too much water can be just as hazardous to an animal cell as losing water.
- ❖ If the cell is immersed in a solution that is **hypotonic** to the cell, water enters the cell faster than it leaves, and the cell swells and lyses (bursts) like an overfilled water balloon.

▲ **Figure 7.12 The water balance of living cells.** How living cells react to changes in the solute concentration of their environment depends on whether or not they have cell walls. **(a)** Animal cells, such as this red blood cell, do not have cell walls. **(b)** Plant cells do. (Arrows indicate net water movement after the cells were first placed in these solutions.)

Plant cell– turgor pressure



- ❖ The cells of plants, prokaryotes, fungi, and some protists are surrounded by walls.
- ❖ A plant cell in a solution hypotonic to the cell contents swells due to osmosis until the elastic cell wall exerts *turgor pressure* on the cell, opposing further water uptake.
 - ✓ *At this point, the cell is **turgid** (very firm), a healthy state for most plant cells.*
 - ✓ *Turgid cells contribute to the mechanical support of the plant.*
- ❖ If a plant cell and its surroundings are isotonic, there is no water movement into the cell. The cell becomes **flaccid** (limp), and the plant may wilt.
- ❖ The cell wall provides no advantages when a plant cell is immersed in a hypertonic solution.
- ❖ As the plant cell loses water, its volume shrinks. Eventually, the plasma membrane pulls away from the wall. This **plasmolysis** is usually lethal.
- ❖ The walled cells of *bacteria and fungi also plasmolyze in hypertonic environments.*

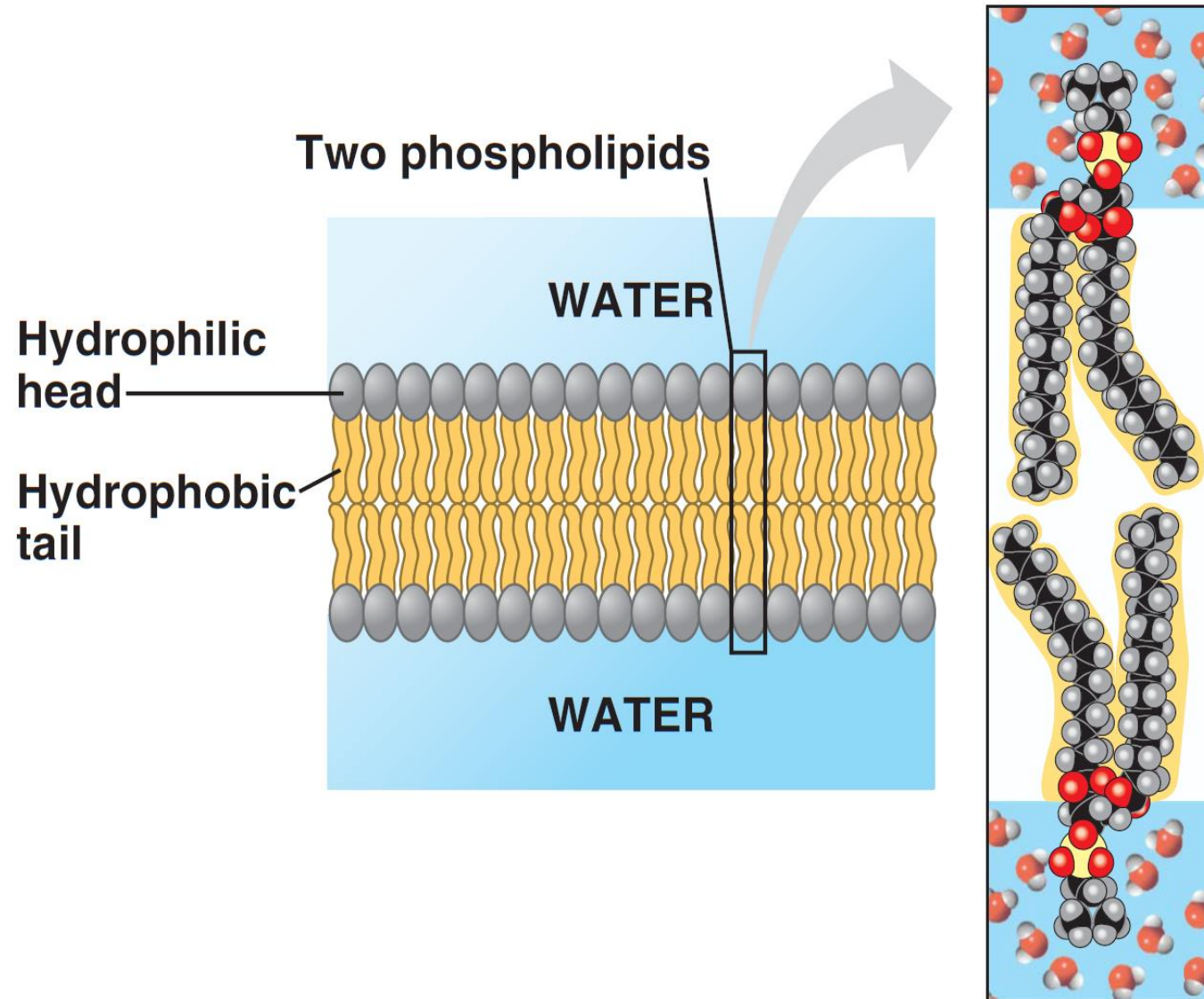
Extras

Extras



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Fluid mosaic model



- ❖ Cellular membranes are **fluid mosaics of lipids and proteins**.
- ❖ The main macromolecules in membranes are **lipids and proteins**, but **carbohydrates** are *also* important.
- ❖ The most abundant lipids are phospholipids.
- ❖ Phospholipids and most other membrane constituents are **amphipathic molecules**, which have both hydrophobic and hydrophilic regions.

Cellular membranes are fluid mosaic of lipid and proteins



- ❖ The arrangement of phospholipids and proteins in biological membranes is described by the **fluid mosaic model**.
 - ✓ In this model, the **membrane is a fluid structure with a “mosaic” of various proteins embedded** in or attached to a double layer (bilayer) of phospholipids.
- ❖ Models of membranes were developed long before membranes were first seen with electron microscopes in the 1950s.
- ❖ In **1915, membranes isolated from red blood cells** were chemically analyzed and found to be composed of lipids and proteins.

Singer and Nicholson model– Freeze fracture techniques and fluid mosaic model



- ✓ In this fluid mosaic model, the hydrophilic regions of proteins and phospholipids are in maximum contact with water, and the hydrophobic regions are in a nonaqueous environment within the membrane.
- ✓ **The membrane is a mosaic of protein molecules bobbing in a fluid bilayer of phospholipids.**
- ❖ A specialized preparation technique, **freeze-fracture**, splits a membrane along the middle of the phospholipid bilayer.
 - ✓ When a freeze-fracture preparation is viewed with an electron microscope, protein particles are interspersed in a smooth matrix, thus supporting the fluid mosaic model.
- ❖ The **fluid mosaic model continues to be refined**. **The summary of it is as follows:**
 - ✓ Membranes may be “**more mosaic than fluid**,” with groups of proteins associated in specialized patches to carry out common functions.
 - ✓ Lipids also appear to form defined regions.
 - ✓ The membrane may also contain more proteins than previously thought.

Cellular membranes are fluid mosaic of lipid and proteins



- ❖ In **1925**, two Dutch **scientists** reasoned that **cell membranes must be phospholipid bilayers**.
 - ✓ The molecules in the bilayer are arranged such that the hydrophobic fatty acid tails are sheltered from water while the hydrophilic phosphate groups interact with water.
- ❖ Actual membranes adhere more strongly to water than do artificial membranes composed only of phospholipids.
- ❖ In **1935**, Hugh Davson and James Danielli proposed a **sandwich model** in which the phospholipid bilayer lies between two layers of globular proteins.
- ❖ Early images from electron microscopes seemed to support the Davson-Danielli model, and until the 1960s, it was widely accepted as the structure of the plasma membrane and internal membranes.

Passive transport: Osmosis



- ❖ The diffusion of water across a selectively permeable membrane is called **osmosis**.
- ❖ The movement of water across cell membranes and the balance of water between the cell and its environment are crucial to organisms.
- ❖ Both solute concentration and membrane permeability affect **tonicity, the ability of a surrounding solution to cause a cell to gain or lose water**.
 - ✓ *The tonicity of a solution depends in part on its concentration of solutes that cannot cross the membrane (nonpenetrating solutes) relative to the concentration of solutes in the cell itself.*
 - ✓ *If there are more nonpenetrating solutes in the surrounding solution, water tends to leave the cell, and vice versa.*
- ❖ If a cell without a cell wall, such as an animal cell, is immersed in an environment that is **isotonic** to the cell, there is no *net* movement of water across the plasma membrane.
 - ✓ Water diffuses across the membrane, but at the same rate in both directions.